

ED ML LoS vs Competitor Gate Engine — One-Pager

Purpose

Give decision-makers a crisp comparison between the **ML-only LoS→occupancy** approach and the **MLP + deterministic gate engine** approach, plus a recommended hybrid.

Snapshot

Your project — ML LoS → Occupancy - Predict **patient-level LoS** (CatBoost) from ED features and **aggregate** to hospital-wide bed forecasts (24–72h).

- Key insight: **error symmetry** in LoS often improves occupancy more than simply lowering MAE.
- Future arrivals filled with **averages**; evaluates hospital-wide occupancy (not ward-specific).

Competitor — MLP + Gate Engine - ML outputs **admission risk + LoS buckets**; a **deterministic gate engine** turns this into **policy-bound actions** (priority, next_action, TTL, reason).

- ML is **advice-only**; gates enforce **safety & explainability**; includes ops panel/nudges, label spec for an actionable target.

Head-to-Head

Dimension	ML LoS → Occupancy	MLP + Gate Engine
Primary output	Beds forecast (H=24/48/72h)	Actionable queue rules (priorities, timers, reasons)
Unit of decision	Hospital level (aggregated)	Patient/task level (operational)
Future arrivals	Constant average fill (calendar effects missing)	Not modeled yet (focus on gates)
Granularity	Hospital-wide; no ward/acuity isolation	Per-patient policy; no occupancy total
Governance	No rule layer; no fairness audit	Read-only ML; deterministic gates + guardrails
Calibration	Not reported; post-hoc symmetrization sims	Calibrator in place; shift-aware fallbacks
Data hygiene	Standard ED features; long-stay truncation	Explicit no-leakage (features ≤ t ₀)

Dimension	ML LoS → Occupancy	MLP + Gate Engine
Ops readiness	Good for capacity planning; lacks ops hooks	Good for day-to-day flow; lacks occupancy

Strengths & Gaps

ML LoS → Occupancy

- + Converts patient predictions into bed needs; quantifies what error properties matter.
- Average-fill arrivals, no ward/acuity split, tail truncation; no rule/governance layer.

MLP + Gate Engine

- + Clear safety contract; policy knobs; explainability; ops panel/nudges.
- No hospital-wide occupancy; some label/calibration maturity still needed.

When to Use Which

- **Bed planning 24–72h out:** choose **ML LoS→occupancy** (add calendar-aware arrivals).
- **Running the ED today (escalations, timers, nudges):** choose **Gate Engine** (ML as calibrated hints).

Recommended Hybrid (Best of Both)

- 1) **Keep the occupancy aggregator;** replace average-fill with **calendar-aware arrivals** (day-of-week/season/case-mix).
- 2) **Adopt the gate engine** as a policy layer: define a clear positive label/time window (Gate_Pos v2), costs, exclusions; keep ML **read-only**.
- 3) **Calibrate & audit:** reliability curves, Brier/ACE, and **fairness slices** (sex, age, acuity, arrival hour, language, insurance).
- 4) **Wire together:** feed **LoS quantiles + admit risk** into gates; publish **ward/acuity/isolation** occupancy for staffing.

Pilot & Metrics

Shadow-mode → capped pilot with governance sign-off. Track: - **Beds MAE@H** (H=24/48/72) per ward + P95, **overflow/shortfall days**, signed bias.

- **Nudge utility:** TP – λ ·FP – μ ·FN; clinician “Not helpful” rate; override counts.
- **Flow:** boarding time, diversion hours, elective cancellations avoided.
- **Fairness:** error/calibration parity across subgroups.

Decision Guidance (TL;DR)

- Need **explainable, policy-controlled actions now** → start with **Gate Engine**.
 - Need **reliable capacity forecasts** → lean on **ML LoS→occupancy** with better arrivals.
 - Need **both** → implement the **hybrid** above; pilot in shadow-mode, then A/B with decision-centric metrics.
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Extended Analysis & Strategy

Engineering Maturity — Verdict

- **More production-ready for per-patient ops:** the gate-engine stack (read-only ML → deterministic rules, invariants, leakage checks, calibration fallback, ops panel, staged rollout).
- **More scientifically novel for capacity:** the LoS→occupancy translator and the finding that **symmetric LoS errors** reduce occupancy bias more than simply shrinking MAE.

Credibility & Conference Framing

- They're only a threat if you present the *same* problem. Keep the split clear: **who to act on now (gates)** vs. **how many/where in 24–72h (capacity)**.
- Be candid about two gaps (average-fill arrivals; artificial timeline) and state fixes. Match their safety posture: **ML advice-only for ops; policy lives in rules**.

If They Add Capacity — Why It's Non-Trivial

- They need a **calendar-aware arrivals model** (DoW/season/holiday + case-mix) to replace averages; otherwise time-dependent error remains high.
- **Aggregation quality ≠ point accuracy:** they must optimize for **symmetric signed errors** and low occupancy bias, not just LoS MAE.
- **Calibration under shift:** turning bucketed probabilities into reliable occupancy distributions with uncertainty bands.
- **Granularity:** ward/acuity/isolation forecasts and translation into **staffable rosters** (skill-mix, blocked beds).
- **Evaluation:** decision-centric metrics (Beds MAE@H, P95 error, overflow/shortfall days, signed bias) and external validation with real timestamps.
- **Fairness & privacy:** subgroup performance, guardrails on sensitive features, synthetic-data privacy tests.

Defensibility Moves (Next 4 Weeks)

1) **Ship arrivals:** calendar-aware arrivals; show delta vs. average-fill on 24/48/72h MAE and signed bias. 2) **Train for aggregation:** quantile/pinball loss; add a symmetry regularizer; report occupancy MAE@H, P95, signed bias, overflow days; add isotonic/quantile calibration. 3) **Go granular:** ward/acuity/isolation forecasts; map to staffing plans and skill-mix; include blocked-bed logic. 4) **Governance parity:** advice-only ML for ops; deterministic gate layer for nudges; reliability curves, fairness slices, drift monitoring, override

logging. 5) **Rare cohorts:** cap synthetic:real ($\leq 1:1$), utility-filter synthetic, run privacy audits; use DP-CTGAN for any external sharing.

Conference Talk Track & Q&A (Scriptable)

- **Open with problem split:** Capacity vs. Ops; show two pipelines and decisions.
- **Claim unique contributions:** LoS→occupancy translator; **error-symmetry** impact on occupancy.
- **Show fixes:** arrivals model replacing averages; ward-level outputs; calibration + fairness dashboards.
- **Safety alignment:** read-only ML for ops; deterministic policy gates.
- **Q&A snippets:**
 - *"Why no gate engine?"* We isolate capacity; for ops we use read-only ML and rules—same safety contract.
 - *"Seasonality/arrivals?"* We replaced averages with calendar-aware arrivals; that's the main driver of time-dependent accuracy.
 - *"Calibration/fairness?"* Reliability curves + subgroup slices; same calibrator-with-fallback pattern for ops surfaces.
 - *"Could they do capacity?"* Yes, with arrivals modeling, aggregation-aware training, calibration, and ward-level translation—this is exactly what we're shipping now.

Bottom Line

Use their **gate engine** for governed, explainable per-patient actions; keep and harden your **capacity module** for reliable 24–72h staffing and beds. Present them as **complementary** and lead with concrete fixes and decision-centric metrics.